



High piezoelectricity and thermal stability of Ce-doped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ -based high-temperature ceramics

Tao Lin^{a,c}, Wen-Ying Fan^b, Chuan-Yang Tao^c, Jun Ma^c, Bin-Jie Chen^{b,*}, Zhong-Shang Dou^b, Bing-Lin Shen^b, Min Ju^b, Jing-Tong Lu^d, Mei-Peng Zhong^{a,*}, Yu-Qing Zhou^{a,c}, Mao-Hua Zhang^b, Wen Gong^{e,*}, Fang-Zhou Yao^{b,f,**}, Ke Wang^{b,d}

^a College of Mechanical and Electrical Engineering, Jiaxing Nanhu University, Jiaxing 314001, China

^b Research Center for Advanced Functional Ceramics, Wuzhen Laboratory, Jiaxing 314500, China

^c College of Mechanical and Electrical Engineering, Wenzhou University, Wenzhou 325035, China

^d State Key Laboratory of New Ceramics and Fine Processing, School of Materials Science and Engineering, Tsinghua University, Beijing 100084, China

^e Tongxiang Tsingfeng Technology Co. Ltd, Jiaxing, Zhejiang 314501, China

^f Center of Advanced Ceramic Materials and Devices, Yangtze Delta Region Institute of Tsinghua University, Zhejiang 314006, China

ARTICLE INFO

Keywords:

$\text{CaBi}_2\text{Nb}_2\text{O}_9$

Piezoelectric properties

Thermal stability

In situ measurement

Temperature-dependent d_{33}

ABSTRACT

As the industry advances at a swift pace, the demand for piezoelectric materials that exhibit both exceptional piezoelectric properties and excellent thermal stability is on the rise. However, it remains a significant challenge to achieve a balance between high piezoelectricity and thermal stability in these materials. In this study, the phase structure, microstructure, and thermal stability of Ce-doped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ (CLBBN-xCe) high-temperature bismuth layer-structured piezoceramics are studied. It is revealed that the CLBBN-0.1Ce ceramics stand out with a piezoelectric coefficient d_{33} of 21.7 pC/N and a high Curie temperature of 907 °C. Furthermore, *in situ* temperature measurement demonstrates that the d_{33} has a change rate of 20.8 % from room temperature to 400 °C. We found that grain size refinement and domain morphology contribute to excellent electrical properties of the Ce-doped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ -based ceramics. The superior performance of these ceramics suggests they hold great promise for high-temperature applications.

1. Introduction

Piezoelectric materials are widely used in sensors, actuators and transducers, due to their ability to realize the mutual conversion of electrical and mechanical energies [1–5]. As industries such as aerospace, nuclear power, and automotive manufacturing progress at a rapid pace, there is a growing need for piezoelectric materials that can operate at higher temperatures and exhibit enhanced thermal stability [6–11]. Piezoelectric coefficient (d_{33}) is one of the most important parameters to measure the piezoelectric properties of materials, and its value is sensitive to temperature fluctuations. When the environmental temperature exceeds the Curie temperature (T_C) of piezoelectric ceramics, the crystal structure will change from an asymmetric configuration to a symmetrical configuration, which leads to the loss of piezoelectric functionality. To maintain the safety and reliability of piezoelectric ceramics, their

operational temperature is usually limited to below 50 % of the Curie temperature, which significantly curtails their potential applications [12–14]. Therefore, the development of piezoelectric ceramics that offer a high piezoelectric coefficient, a high Curie temperature, and superior thermal stability has emerged as a key research objective in the quest for high-temperature piezoelectric ceramics.

Currently, the commercial applications of BLSFs are predominantly focused on ceramics based on $\text{Bi}_4\text{Ti}_3\text{O}_{12}$, which features a three-layer structure, and $\text{CaBi}_4\text{Ti}_4\text{O}_{15}$, with a four-layer configuration. These materials are primarily utilized in applications that require operation temperatures of up to 260 °C and 482 °C, respectively, highlighting their tailored performance for specific thermal conditions. Bismuth-layered ferroelectrics, represented by $\text{CaBi}_2\text{Nb}_2\text{O}_9$ (CBN), are promising candidates for high-temperature piezoelectric applications due to their exceptional thermal stability and elevated Curie temperatures, but they

* Corresponding authors.

** Corresponding author at: Research Center for Advanced Functional Ceramics, Wuzhen Laboratory, Jiaxing 314500, China.

E-mail addresses: chenbj@wuzhenlab.com (B.-J. Chen), zhongmeipeng@jxnhu.edu.cn (M.-P. Zhong), pgong@139.com (W. Gong), yaofangzhou@xjtu.edu.cn (F.-Z. Yao).

<https://doi.org/10.1016/j.jeurceramsoc.2024.116955>

Received 1 July 2024; Received in revised form 23 September 2024; Accepted 27 September 2024

Available online 29 September 2024

0955-2219/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

have relatively low intrinsic piezoelectric properties. To enhance the performance of CBN ceramics for wider high-temperature applications, numerous modification strategies have been explored. For example, Liu et al. [15] successfully enhanced the piezoelectricity of CBN ceramics through (Na, Ce) modification, significantly enhanced the thermal stability of the lattice by reducing the rotation of the Nb-O octahedron during thermal depoling process, which is still maintaining 92 % of the initial d_{33} after depoling at 600 °C. Shen et al. [16] fabricated a Ce and W co-doped $\text{Ca}_{0.96}\text{Ce}_{0.04}\text{Bi}_2\text{Nb}_{2-x}\text{W}_x\text{O}_9$ ceramic using conventional solid-state sintering method, achieving a d_{33} of 13.4 pC/N and a resistivity ρ of $3.7 \times 10^9 \Omega \text{ cm}$ at 500 °C, along with remarkable thermal stability up to 800 °C. Wang et al. [17] reported on Cu/W co-doped $\text{Ca}_2\text{Bi}_2\text{Nb}_{2-x}(\text{Cu}_{1/4}\text{W}_{3/4})_x\text{O}_9$ high Curie temperature piezoelectric ceramics, exhibiting a d_{33} of 13.8 pC/N and a Curie temperature of 918 °C. These ceramics demonstrated commendable thermal stability, retaining a d_{33} of 12.7 pC/N after annealing at 900 °C for 30 mins. Luo et al. [18] improved the piezoelectric properties and thermal stability of CBN-based ceramics by Na^+ and Sm^{3+} co-doping, yielding a d_{33} of 12.8 pC/N and a Curie temperature of 949 °C. After annealing treatment at 900 °C for 2 h, the d_{33} value of the materials remained at approximately 90 % of its initial value. Despite these significant advances, the majority of current researches on the thermal stability of $\text{CaBi}_2\text{Nb}_2\text{O}_9$ based ceramics concentrated on the *ex situ* high temperature stability (annealing at high temperature for 1–2 h) [19–22]. The d_{33} values are measured after high-temperature annealing, which may not accurately represent the d_{33} values at actual high temperature environment. There is a scarcity of studies that apply *in situ* temperature-dependent d_{33} measurement to $\text{CaBi}_2\text{Nb}_2\text{O}_9$ based ceramics.

In this work, we improve the electrical properties of CBN-based piezoelectric ceramics by Ce doping replaces Bi ions in $(\text{Bi}_2\text{O}_2)^{2+}$ on the basis of replacing Ca with Li/Bi at A-site. The resulting $\text{Ca}_{0.6}(\text{Li}_{0.5}\text{Bi}_{0.5})_{0.4}\text{Bi}_{2-0.4x}\text{Ce}_{0.4x}\text{Nb}_2\text{O}_9$ (CLBBN-xCe) ceramic exhibits a high d_{33} value of 21.7 pC/N and a high Curie temperature of 907 °C, substantially outperforming the pristine CBN ceramics in terms of piezoelectric response. In addition, *in situ* temperature-dependent measurements demonstrate that the d_{33} value of the CLBBN-0.1Ce ceramics at 420 °C retains 90 % of its value at room temperature, underscoring its exceptional piezoelectric properties and thermal stability. We conducted in-depth analyses to elucidate the interplay between phase structure, microstructure, piezoelectric properties, and thermal stability. This investigation unveils the effect of Ce-doping on the significant improving piezoelectric properties and maintaining high thermal stability, offering valuable insights for the development of advanced piezoelectric materials capable of operating in high-temperature environments.

2. Experimental procedure

Samples with nominal compositions of $\text{Ca}_{0.6}(\text{Li}_{0.5}\text{Bi}_{0.5})_{0.4}\text{Bi}_{2-0.4x}\text{Ce}_{0.4x}\text{Nb}_2\text{O}_9$ (CLBBN-xCe, $x = 0, 0.05, 0.1, 0.15, 0.2$) high-temperature ceramics were synthesized by the solid-state reaction. High-purity raw materials including Ca_2CO_3 (99 %), Li_2CO_3 (99.9 %), CeO_2 (99.95 %), Bi_2O_3 (99 %), Nb_2O_5 (99.5 %) were used. Each constituent was meticulously weighed according to the stoichiometric ratios and then subjected to a thorough ball-milling process for a continuous period of 24 h. Subsequently, the milled powders were calcined at 850 °C for 3 h. The calcined powders were granulated and pressed into disks with 8.5 mm in diameter and 2 mm in thickness. These green compacts were sintered within a sealed crucible at a high temperature of 1120 °C for a period of 4 h.

The crystal structures of the samples were characterized using X-ray diffraction (XRD, D8 ADVANCE, Bruker, Germany). The thermally-etched surface of the samples was examined by scanning electron microscopy (SEM, SIGMA 300, ZEISS, Germany). The ferroelectric properties of the samples were analyzed using a ferroelectric analyzer (TF Analyzer 2000E, aixACCT, Germany). The samples were poled under a

DC field of 15 kV/mm in silicone oil at 160 °C for about 30 min (ET2673D, ENTAI, China). The room-temperature d_{33} of the samples were determined using a quasi-static d_{33} meter (ZJ-3AN, Chinese Academy of Sciences, China). The *in situ* temperature-dependent d_{33} of the samples were measured by the *in situ* temperature-dependent d_{33} measurement device (TZFD-600, Julang Technology, China). The temperature-dependent Raman spectra of the samples were characterized using Raman spectroscopy (LabRAM HR800, HORIBA, France) under the condition of 100–1000 cm^{-1} , and over a temperature gradient from 25 to 500 °C. The temperature-dependent of the dielectric permittivity and the dielectric loss was measured by precision impedance analyzer (Agilent HP4294A, KEYSIGHT, America), operating at a frequency of 1 MHz, across the temperature spectrum of 25–1000 °C. The nanoscale domain morphologies of the samples were captured using a piezoresponse force microscope (Jupiter XR, Oxford Instruments, Britain).

3. Results and discussion

Fig. 1(a) shows the room temperature XRD patterns of CLBBN-xCe ceramics in the 2θ range of 20° to 70°. The observed diffraction peaks are indexed in accordance with the JCPDS-49-0608 ($\text{CaBi}_2\text{Nb}_2\text{O}_9$), revealing that all samples and the original component samples have the same orthorhombic structure. The absence of any undesired secondary phase substantiates the homogeneous incorporation of cerium (Ce) into the host lattice. Notably, the peak with highest intensity among the XRD patterns is identified as the (115) reflection, which corresponds to the theory that the most intense diffraction peak of the Aurivillius phase is the (112m+1) [23–26]. As shown in Fig. 1(b), the crystal structure evolution of the samples was further analyzed through the detailed analysis of the profiles near 29.5° and 33°. It is observed that as x increases, the (115) peak progressively shifts toward higher 2θ angles, indicating a diminution in the lattice parameter with the increasing concentration of the Ce dopant. The Fig. 1(c) shows that the (200) and (020) peaks progressively coalesce with the incremental substitution of Ce, suggesting that the doping of Ce leads to the appearance of pseudo-tetragonal phase. The pseudo-tetragonal phase is conducive to polarization switching, and improves the spontaneous polarization (P_s) of the CLBBN-xCe ceramics.

In order to further analyze the influence of Ce-doping on the structure of CLBBN ceramics, Rietveld refinement for XRD profiles was conducted using the $A2_1am$ space group. Fig. 2(a)–(b) display the Rietveld refinement results for the CLBBN-xCe ($x = 0$ and 0.1) ceramics, respectively, which exhibit a high degree of agreement with the original XRD patterns. The refinement factors wR_p , R_p and χ^2 are all within an acceptable range. Fig. 2(c) shows the lattice parameters of CLBBN-xCe. There is a discernible trend of reduction in the lattice parameters a and c

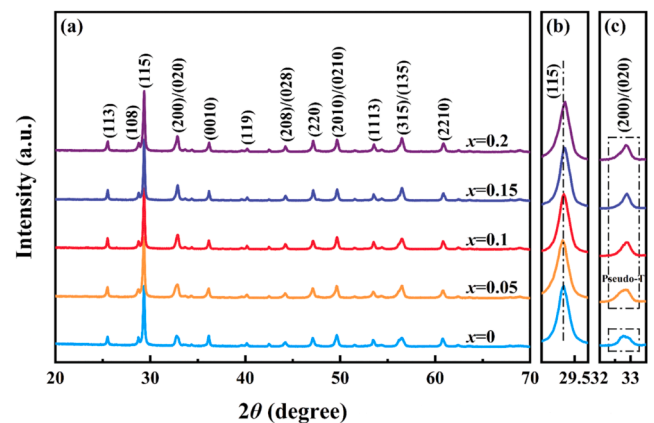


Fig. 1. (a) XRD patterns of CLBBN-xCe ceramics; enlarged (b) (115) and (c) (200)/(020) diffraction peaks.

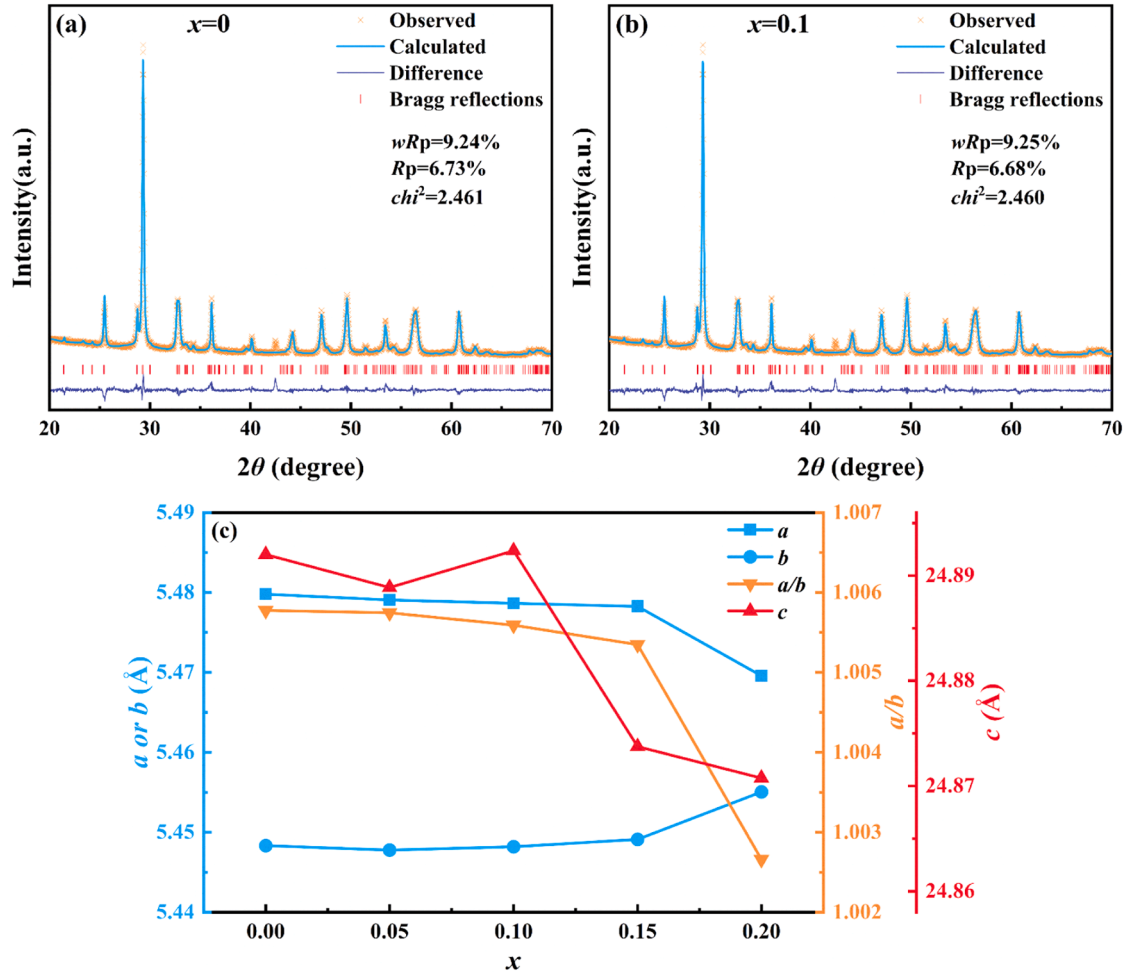


Fig. 2. (a)-(b) Rietveld refinements of CLBBN-xCe ($x = 0$ and 0.1) ceramics; (c) Lattice parameters a , b , c and a/b values of CLBBN-xCe ceramics.

as the dopant concentration x escalates, while the lattice parameter b shows an opposite trend. The degree of orthogonal distortion can be expressed as the value of a/b [27]. As the Ce content increases, the a/b ratio declines, denoting an alleviation of orthogonal distortion whereas an intensification of pseudo-tetragonal distortion, which is consistent with the results that the (200) and (020) peaks are gradually merged in XRD patterns.

Fig. 3(a)–(e) show the scanning electron microscopy (SEM) images of the thermally-etched CLBBN-xCe ceramics. The observation reveals a plate-like structure characteristic of bismuth layer-structured piezoelectric ceramics. This morphology is attributed to the enhanced grain growth rate within the a - b plane [28–30]. Fig. 3(f) shows the grain size of the CLBBN-xCe ceramics as a function of the Ce dopant concentration x . The grain size initially diminishes and then increases with increasing x . Specifically, at the Ce doping levels of 0.1 and 0.15, the grain sizes are 5.16 μm and 4.13 μm , and the densities are 7.29 and 7.31 g/cm^3 , respectively. These results reveal that Ce-doping effectively refines the grain size and enhances the density of the ceramics, thereby potentially improving their piezoelectric properties and ferroelectric properties. Fig. 3(g)–(i) display the energy dispersive spectroscopy (EDS) mapping of a thermally-etched CLBBN-0.1Ce ceramic. The uniform distribution of all elements can be observed, which is consistent with the results of XRD, and further confirms the diffusion of Ce ions into the CLBBN lattice. This uniform dispersion is crucial for the consistent performance and enhanced functionality of the piezoelectric materials.

Fig. 4(a)–(b) respectively show the polarization-electric field (P - E) hysteresis loops and the corresponding current-electric field (I - E) curves for the CLBBN-xCe ($x = 0$ and 0.1) ceramics, measured under a

frequency of 1 Hz and the temperature of 180 $^{\circ}\text{C}$. Notably, the P - E loop for the CLBBN-0.1Ce ceramic exhibits a more pronounced saturation compared to the CLBBN ceramic. Upon Ce doping, the maximum electric field endurance of the samples is increased from 15 kV/mm to over 21 kV/mm . This phenomenon is potentially attributed to the enhanced density of Ce-doped samples relative to their undoped counterparts. Concurrently, the remnant polarization P_r increases from 9.67 $\mu\text{C}/\text{cm}^2$ to 30.89 $\mu\text{C}/\text{cm}^2$, showing the ceramic's excellent ferroelectric characteristics. These results indicate that an optimal concentration of Ce-doping can significantly improve the ferroelectric properties of CLBBN ceramics. Furthermore, Fig. 4(c)–(d) show the hysteresis loop P - E and current curve I - E of CLBBN-0.1Ce ceramics under different electric fields, respectively. The ferroelectricity of the ceramic is more clearly demonstrated at electric fields exceeding 150 kV/cm , which aligns with the requisite high electric field for sample polarization.

Fig. 5(a) shows the composition-dependent d_{33} values of the CLBBN-xCe ceramics at room temperature. With the increase of Ce dopant concentration, the d_{33} value declines first, then increases to a peak at $x = 0.1$, reaching a high value of 21.7 pC/N , which is among the forefront of CBN-based high-temperature piezoelectric ceramics. The excellent piezoelectric properties were primarily attributed to the improvements of ferroelectric properties, and the facilitation of domain switching of CLBBN-xCe ceramics. In addition, the incorporation of Ce atoms causes the distortion of Nb-O octahedron, resulting in the increase of theoretical spontaneous polarization P_s , which is conducive to the enhancement of polarization. Therefore, these various factors synergistically contribute to the high piezoelectric properties of CLBBN-xCe ceramics. Thermal stability is an important characteristic of the material for high-

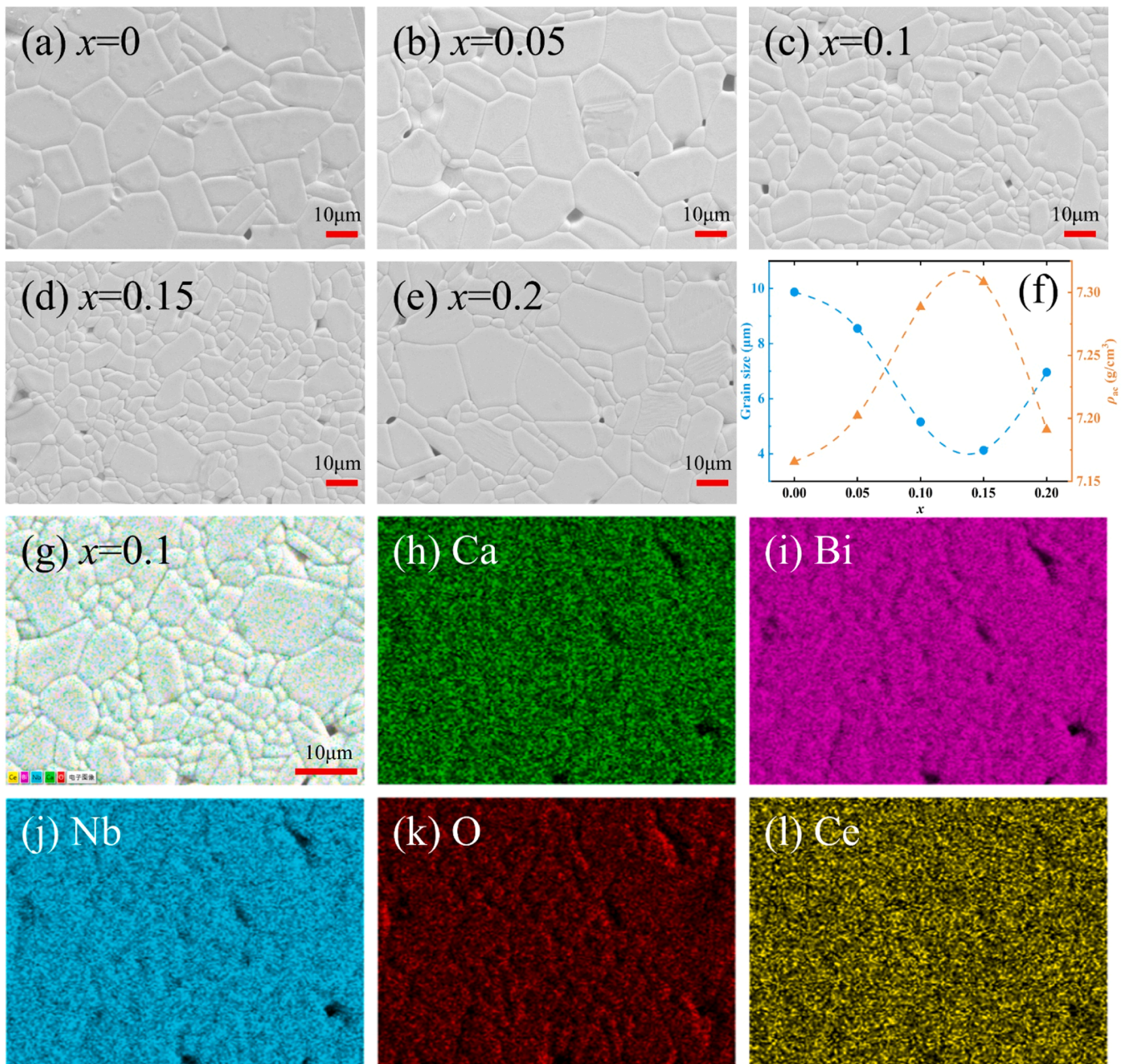


Fig. 3. (a)–(e) SEM images of thermally-etched CLBNN- x Ce ceramic samples; (f) the average grain size and real density of CLBNN- x Ce ceramics; (g)–(l) elemental mapping of the CLBNN-0.1Ce ceramic.

temperature applications [31–33]. In order to display the temperature-dependent d_{33} of these ceramics in real time, the *in situ* temperature-dependent d_{33} of CLBNN- x Ce ($x = 0, 0.1$) were measured. As depicted in Fig. 5(b), the d_{33} values of these ceramics exhibit a trend of initial increase, followed by a gradual stabilization, and ultimately a decline as temperature rises. Raising temperature may increase the mobility of the domain walls of CLBNN ceramic, thereby the extrinsic contribution results in increased piezoelectric coefficient. Notably, at 350 °C, the d_{33} of CLBNN-0.1Ce ceramics approaches 25 pC/N, which is approximately twice that of the undoped CLBNN ceramics. From room temperature to 400 °C, the rate of change of d_{33} of CLBNN-0.1Ce samples and CLBNN samples were 20.8 % and 19.7 %, respectively. However, it is observed that the depolarization temperature (450 °C) for the CLBNN-0.1Ce ceramics is lower, compared to 500 °C for the CLBNN ceramics. In BLSFs, the $(\text{Bi}_2\text{O}_3)^{2+}$ layer acts as an insulating slab, the incorporation of Ce into $(\text{Bi}_2\text{O}_3)^{2+}$ layer leads to a decrease in the

resistivity of CLBNN-0.1Ce ceramics. In addition, Ce ions can readily change their oxidation states between +3 and +4, if Ce^{4+} replaces Nb^{5+} in the B-site, oxygen vacancies will be generated. And when the temperature reaches up to 400 °C, the activation energy of charge carrier (such as oxygen vacancies) decreases rapidly, the mobility of the oxygen vacancy increases, and the resistivity further decreases as the temperature increases, resulting in a decrease in the electrical properties of the ceramics at temperatures much lower than T_C [27,34]. Correlation between the d_{33} and T_C of previously reported CBN-based ceramics is shown in Fig. 5(c) [11,15–18,20,23,24,26,31,38,40,43]. These findings indicate that an optimal Ce-doping concentration substantially boosts the piezoelectric properties of CLBNN ceramics, although this results in a slightly higher sensitivity to temperature variations in their piezoelectric performance.

To delve deeper into the phase-transition behavior of the Ce-doped CLBNN-based ceramics at high temperature, temperature dependence

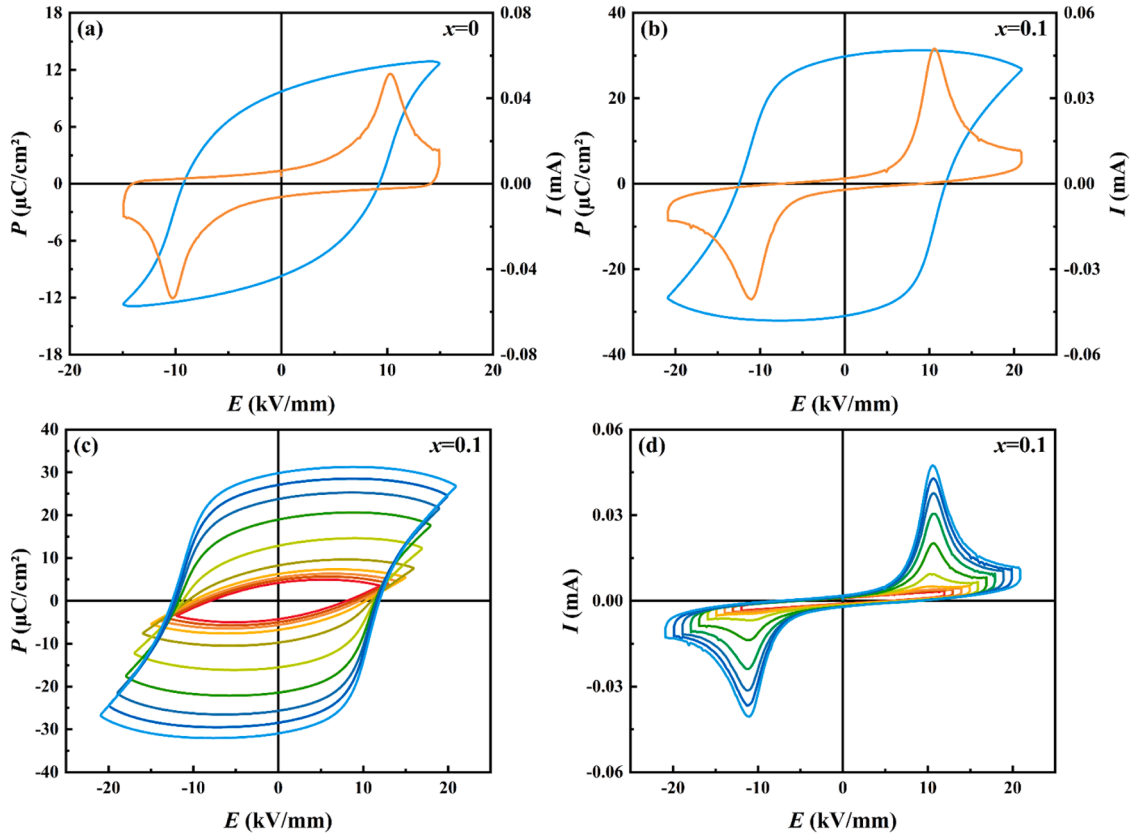


Fig. 4. (a) P-E and I-E loops of the CLBBN ceramics; (b) P-E and I-E loops of the CLBBN-0.1Ce ceramics; (c) P-E loops of the CLBBN-0.1Ce ceramics at different electric fields; (d) I-E loops of the CLBBN-0.1Ce ceramics at different electric fields.

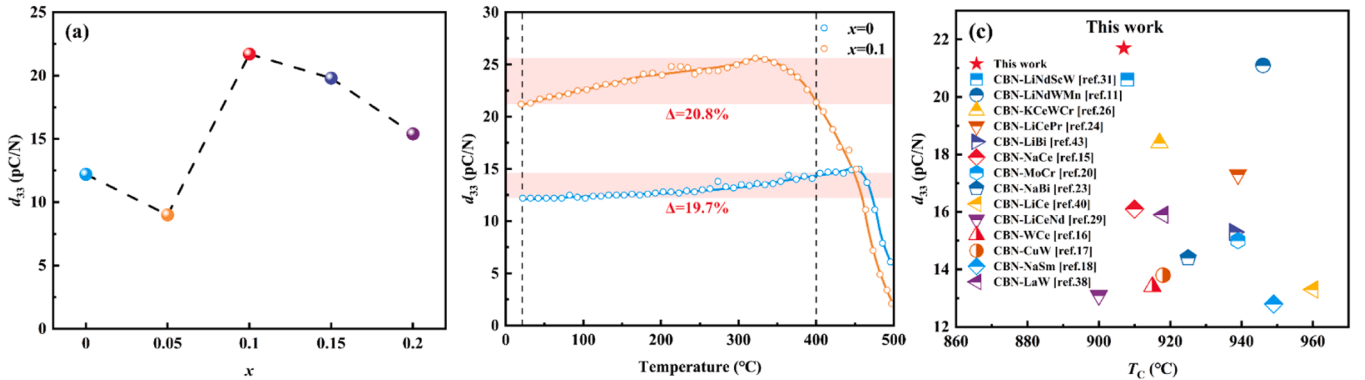


Fig. 5. (a) The room temperature d_{33} values of the CLBBN-xCe ceramics at; (b) *In situ* temperature-dependent d_{33} of the CLBBN-xCe ($x = 0$ and 0.1) ceramics. The solid lines are for an eye guide. (c) A summary of literature on T_c and d_{33} correlation of CBN-based ceramics.

of Raman spectra for the samples was characterized [35]. Fig. 6(a)-(b) show the temperature dependence of Raman spectra for the representative CLBBN-xCe ($x = 0$ and 0.1) samples, recorded within the wave-number range of $100\text{--}1000\text{ cm}^{-1}$ and across a temperature span of $25\text{--}500\text{ }^{\circ}\text{C}$. All the samples show the same phonon modes at about 189 , 213 , 597 , 822 cm^{-1} . As temperature rises, there is a general attenuation in the intensity of the Raman peaks, accompanied by a progressive reduction in the number of discernible peaks in the Raman spectra. Fig. 6(c)-(f) focus on the specific modes at 189 cm^{-1} (denoted by ν_2), 213 cm^{-1} (ν_3), 597 cm^{-1} (ν_7), and 822 cm^{-1} (ν_9). It can be seen that the ν_2 and ν_3 modes exhibit a slight lower-frequency shift after Ce-doping, which is indicative of alterations in the vibrational modes of the A-site cations [36–38]. Moreover, the mode at 597 cm^{-1} (ν_7) shifts significantly to a lower frequency after Ce-doping, suggesting a reduction in

vibration of the NbO_6 octahedron. The mode at 822 cm^{-1} (ν_9) is correlated with the structural symmetry of the material.

It is evident that as the temperature increases, the ν_2 , ν_3 , and ν_7 modes progressively downshift to lower frequencies. The variations in the ν_2 and ν_3 modes imply that the vibrations of the A-site ions are influenced by the temperature. The shifts in the ν_7 and ν_9 modes suggest that temperature-induced structural alterations are mediated through changes in the Nb-O bond lengths [39]. Observations indicate that beyond $350\text{ }^{\circ}\text{C}$, the alterations in the modes of the CLBBN-0.1Ce ceramics are markedly subdued. Concurrently, the *in situ* temperature-dependent d_{33} measurements reveal a notable decrement in the d_{33} values starting at approximately $350\text{ }^{\circ}\text{C}$. This correlation suggests a linkage between the attenuation of modal variations and the decline in piezoelectric performance, pinpointing $350\text{ }^{\circ}\text{C}$ as a critical

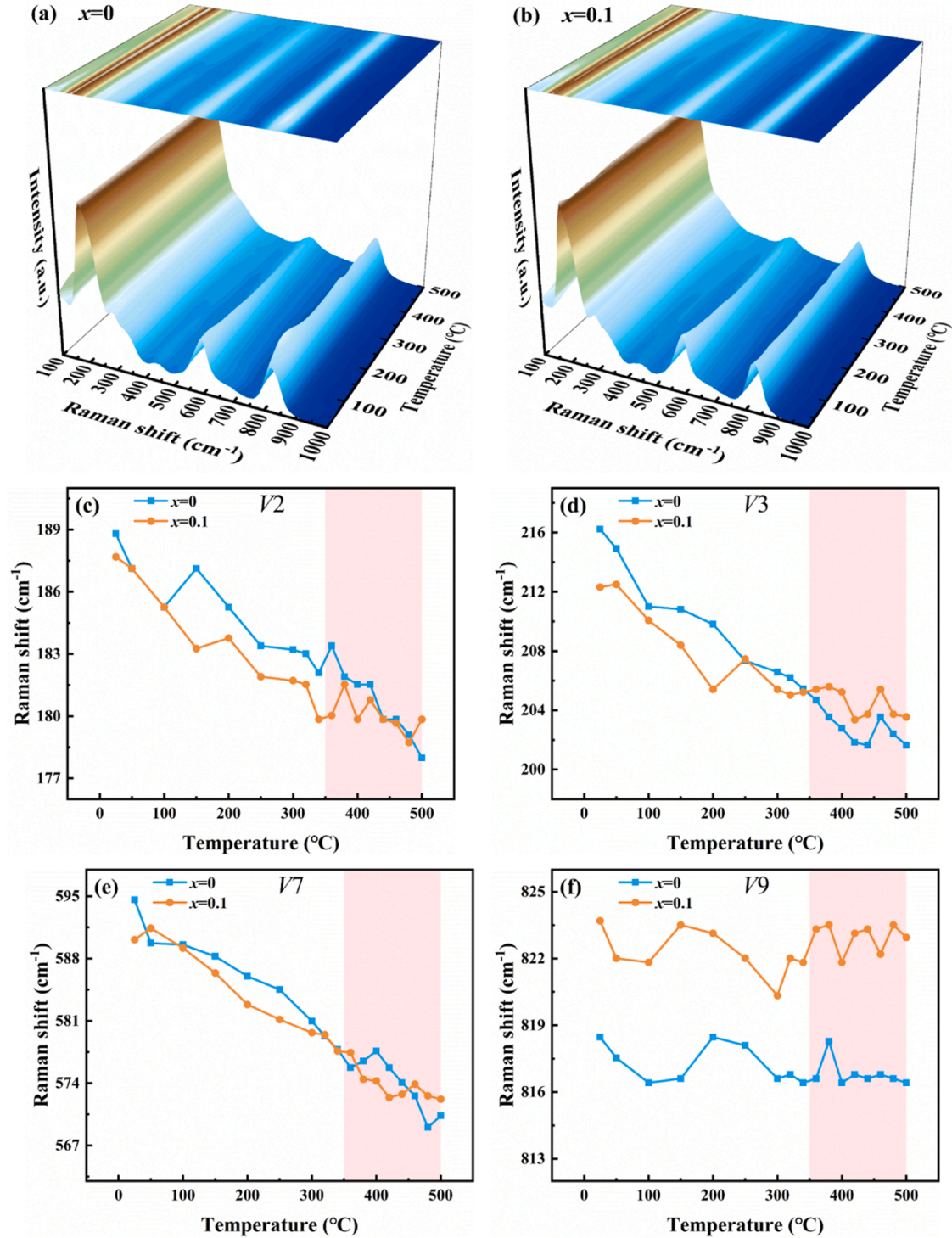


Fig. 6. (a)-(b) The temperature dependence Raman spectra of CLBBN- x Ce ($x = 0$ and 0.1) ceramics; (c)-(f) The variation of each Raman peak with temperature.

threshold where significant changes in the material's properties are observed.

Fig. 7(a) shows the temperature dependence of the dielectric constant ϵ_r of the CLBBN- x Ce ($x = 0$ and 0.1) ceramics measured at a frequency of 1 MHz. The dielectric constant ϵ_r of both ceramics exhibits a pronounced peak at the Curie temperature, signifying the critical phase transition from ferroelectric phase to paraelectric phase [40,41]. Notably, the Curie temperature of the CLBBN-0.1Ce ceramics (907 °C) is slightly lower than that of the CLBBN ceramics (920 °C). This slight decrease is attributed to the increased pseudo-tetragonal distortion after

Ce doping, resulting in the decrease of phase transition temperature [42–44]. In addition, For BLSFs families, oxygen vacancies will be generated due to the Bi-loss during the sintering process. Therefore, a weak dielectric anomaly can be observed in the range of 600–800 °C, which may be associated with the defect-related dielectric response. Fig. 7(b) presents the temperature dependence of the dielectric loss $\tan\delta$ of the CLBBN- x Ce ($x = 0$ and 0.1) ceramics, also measured at 1 MHz. At high temperature, the dielectric loss $\tan\delta$ of the CLBBN-0.1Ce ceramics is slightly higher than that of the CLBBN ceramics. However, it is noteworthy that the dielectric loss $\tan\delta$ for both ceramics does not exhibit a

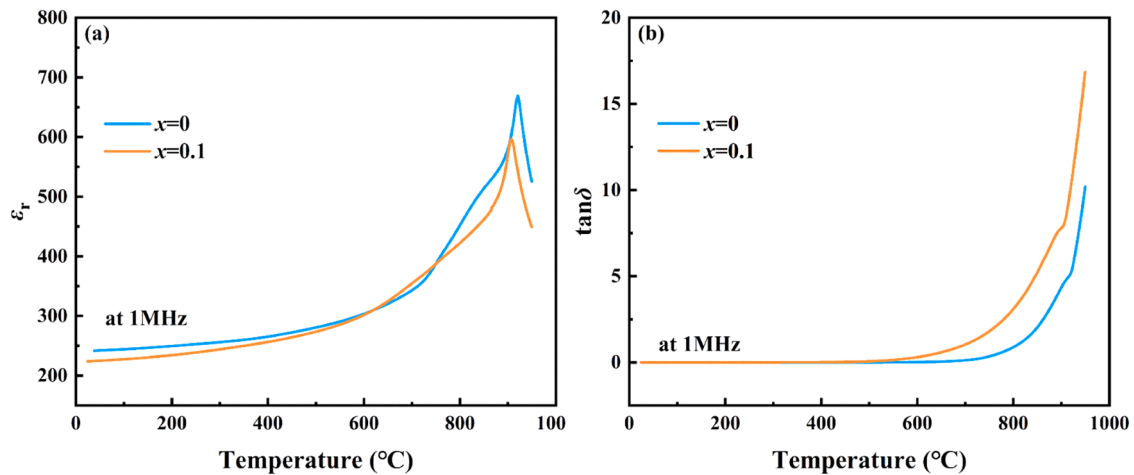


Fig. 7. (a) Temperature dependence of the dielectric constant of CLBBN- x Ce ceramics at 1 MHz; (b) Temperature dependence of the dielectric loss of CLBBN- x Ce ceramics at 1 MHz.

significant increase up to a temperature of 450 $^{\circ}\text{C}$. The dielectric loss $\tan\delta$ at 450 $^{\circ}\text{C}$ is just 0.025 for the CLBBN-0.1Ce ceramic. It also shows a strong thermal stability of the CLBBN-0.1Ce samples, which is an important prerequisite for the application of ceramics at high temperatures. These findings collectively suggest that the Ce-doped CLBBN ceramics have substantial potential for high-temperature applications.

In order to further explore the origins of the excellent properties exhibited by Ce-doped CLBBN samples, an investigation into the domain morphologies was undertaken. Fig. 8(a)–(b) display the out-of-plane phase images of CLBBN- x Ce ($x = 0$ and 0.1) ceramics, as captured by piezoelectric response force microscopy (PFM) at room temperature. A comparison reveals that the CLBBN-0.1Ce ceramic possesses a domain structure with denser and more finely delineated stripe domains compared to the CLBBN ceramic. The piezoelectric effect's principal contributions are twofold: the intrinsic contribution, which arises from field-induced lattice distortions, and the extrinsic contribution, including the dynamics of domain switching and domain wall motion. The finer domain structures, characterized by lower domain wall energy, are more amenable to external stimuli, which is believed to be one of the contributing factors to the excellent piezoelectric properties of the CLBBN-0.1Ce ceramics [45].

4. Conclusions

The CLBBN- x Ce ceramics, with varying Ce concentrations ($x = 0$,

0.05, 0.1, 0.15, and 0.2), were synthesized by the solid-state reaction method. The systematical study results reveal the impact of Ce-doping on the electric properties and thermal stability of CBN-based ceramics. It is noteworthy that the CLBBN-0.1Ce ceramic exhibits a maximum d_{33} value of 21.7 pC/N, complemented by a high Curie temperature of 907 $^{\circ}\text{C}$. In the *in situ* temperature-dependent d_{33} measurements, from room temperature to 400 $^{\circ}\text{C}$, the rate of change of d_{33} of CLBBN-0.1Ce samples is 20.8 %, which is also an important attribute for practical applications. The results of this study highlight the significant advantages of the Ce-doped CLBBN ceramics, suggesting their promising potential for application in high-temperature environments. These ceramics stand out as robust candidates for applications where both high thermal stability and superior piezoelectric properties are concurrently demanded.

CRediT authorship contribution statement

Min Ju: Writing – review & editing. **Bing-Lin Shen:** Data curation. **Jing-Tong Lu:** Data curation. **Yu-Qing Zhou:** Writing – review & editing. **Mei-Peng Zhong:** Writing – review & editing. **Wen Gong:** Writing – review & editing. **Wen-Ying Fan:** Data curation. **Mao-Hua Zhang:** Writing – review & editing. **Tao Lin:** Writing – original draft. **Ke Wang:** Writing – review & editing. **Jun Ma:** Data curation. **Fang-Zhou Yao:** Writing – review & editing. **Chuan-Yang Tao:** Data curation. **Zhong-Shang Dou:** Writing – review & editing. **Bin-Jie Chen:** Writing – review & editing.

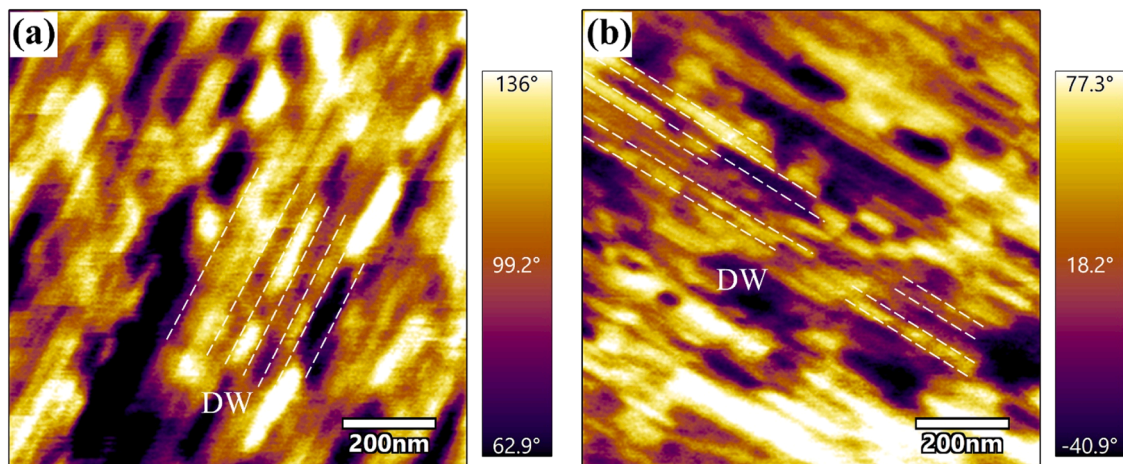


Fig. 8. (a) The out-of-plane phase images of CLBBN ceramics; (b) The out-of-plane phase images of CLBBN-0.1Ce ceramics.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by National Nature Science Foundation of China (No. 52325204, 52032005, U22A20254), Key R&D Program of Zhejiang (No. 2022C01229), and the Program for Jiaxing Leading Innovative and Entrepreneurial Teams.

References

- [1] B. Jaffe, W. Cook, H. Jaffe, *Piezoelectric Ceramics*, Academic Press, London, 1971.
- [2] H. Jaffe, D.A. Berlincourt, *Piezoelectric transducer materials*, *Proc. IEEE* 53 (10) (1965) 1372–1386.
- [3] R.C. Turner, P.A. Fuierer, R.E. Newnham, T.R. Shrout, *Materials for high temperature acoustic and vibration sensors: a review*, *Appl. Acoust.* 41 (4) (1994) 299–324.
- [4] S. Zhang, F. Yu, *Piezoelectric materials for high temperature sensors*, *J. Am. Ceram. Soc.* 94 (10) (2011) 3153–3170.
- [5] M. Zhang, C. Shen, C. Zhao, M. Dai, F. Yao, B. Wu, J. Ma, H. Nan, D. Wang, Q. Yuan, L.L.D. Silva, L. Fulanvic, A. Schokel, P. Liu, H. Zhang, J. Li, N. Zhang, K. Wang, J. Rodel, M. Hinterstein, *Deciphering the phase transition-induced ultrahigh piezoresponse in (K, Na) NbO₃-based piezoceramics*, *Nat. Commun.* 13 (1) (2022) 3434.
- [6] A. Moure, A. Castro, L. Pardo, *Aurivillius-type ceramics, a class of high temperature piezoelectric materials: drawbacks, advantages and trends*, *Prog. Solid State Chem.* 37 (1) (2009) 15–39.
- [7] S. Huang, J. Zeng, L. Zheng, Z. Man, X. Ruan, X. Shi, G. Li, *A novel piezoelectric ceramic with high Curie temperature and high piezoelectric coefficient*, *Ceram. Int.* 46 (5) (2020) 6212–6216.
- [8] X. Xie, Z. Zhou, R. Liang, X. Dong, *Significantly enhanced piezoelectric performance in Bi₄Ti₃O₁₂-based high-temperature piezoceramics via oxygen vacancy defects tailoring*, *J. Mater.* 7 (1) (2021) 59–68.
- [9] M. Zhang, K. Wang, Y.J. Du, G. Dai, W. Sun, G. Li, D. Hu, H. Thong, C. Zhao, X. Xi, Z. Yue, J. Li, *High and temperature-insensitive piezoelectric strain in alkali niobate lead-free perovskite*, *J. Am. Chem. Soc.* 139 (10) (2017) 3889–3895.
- [10] Y. Dong, K. Zou, R. Liang, Z. Zhou, *Review of BiScO₃-PbTiO₃ piezoelectric materials for high temperature applications: fundamental, progress, and perspective*, *Prog. Mater. Sci.* 132 (2023) 101026.
- [11] C. Long, B. Wang, W. Ren, K. Zheng, H. Fan, D. Wang, L. Liu, *Significantly enhanced electrical properties in CaBi₂Nb₂O₉-based high-temperature piezoelectric ceramics*, *Appl. Phys. Lett.* 117 (3) (2020) 032902.
- [12] H. Yan, H. Zhang, R. Ueb, M.J. Reece, J. Liu, Z. Shen, Z. Zhang, *A lead-free high-curie-point ferroelectric ceramic, CaBi₂Nb₂O₉*, *Adv. Mater.* 17 (10) (2005) 1261–1265.
- [13] C. Wang, S. Zhang, J. Wang, M. Zhao, C. Wang, *Electromechanical properties of calcium bismuth niobate (CaBi₂Nb₂O₉) ceramics at elevated temperature*, *Mater. Chem. Phys.* 118 (1) (2009) 21–24.
- [14] Y. Li, Z. Zhou, R. Liang, B. Gao, Z. Zhou, X. Dong, *A simple Bi³⁺ self-doping strategy constructing pseudo-tetragonal phase boundary to enhance electrical properties in CaBi₂Nb₂O₉ high-temperature piezoceramics*, *J. Eur. Ceram. Soc.* 42 (6) (2022) 2772–2780.
- [15] G. Liu, S. Ren, C. Wu, D. Wang, F. Li, J. Wu, Q. Chen, *Enhanced thermal stability of (NaCe)-multidoped CaBi₂Nb₂O₉ by A-site vacancies-induced pseudo-tetragonal distortion*, *J. Am. Ceram. Soc.* 101 (10) (2018) 4615–4626.
- [16] Z. Shen, C. Qin, W. Luo, F. Song, Z. Wang, Y. Li, S. Zhang, *Ce and W co-doped CaBi₂Nb₂O₉ with enhanced piezoelectric constant and electrical resistivity at high temperature*, *J. Mater.* 6 (3) (2020) 459–466.
- [17] H. Wang, C. Chen, X. Jiang, X. Huang, X. Nie, L. Huang, X. Yu, *Cu/W Co-doped CaBi₂Nb₂O₉ piezoelectric ceramics on structural and electrical properties*, *J. Alloy. Compd.* 896 (2022) 163078.
- [18] X. Luo, Z. Yan, H. Luo, X. Zhou, B. Li, M. Zhang, D. Zhang, *Greatly improved piezoelectricity and thermal stability of (Na, Sm) Co-doped CaBi₂Nb₂O₉ ceramics*, *Adv. Powder Mater.* 2 (3) (2023) 100116.
- [19] S. Huang, X. Jiang, C. Chen, X. Nie, X. Huang, H. Wang, F. Ye, H. Huang, *Structure and thermal stability investigations of (1-x)CaBi₂Nb₂O₉-xNa_{0.5}Bi_{2.5}Nb₂O₉ ceramics*, *Ceram. Int.* 49 (11) (2023) 17961–17968.
- [20] Z. Chen, Y. Zhang, P. Huang, X. Li, J. Du, W. Bai, L. Li, F. Wen, P. Zheng, W. Wu, L. Zheng, *Enhanced piezoelectric properties and thermal stability in Mo/Cr co-doped CaBi₂Nb₂O₉ high-temperature piezoelectric ceramics*, *J. Phys. Chem. Solids* 136 (2020) 109195.
- [21] X. Zhang, H. Yan, M.J. Reece, *Effect of A site substitution on the properties of CaBi₂Nb₂O₉ ferroelectric ceramics*, *J. Am. Ceram. Soc.* 91 (9) (2008) 2928–2932.
- [22] N. Jabeen, A. Hussain, S. Nawaz, M.A. Qaiser, R. Zafar, S. Ishtiaq, F. Ahmed, U. Masud, T. Sultan, S.T. Muntaha, *Thermally stable piezoelectric performance of MnO₂ inserted pseudo-tetragonal phase existent CaBi₂Nb₂O₉-based ceramics*, *Mater. Technol.* 37 (13) (2022) 2702–2710.
- [23] X. Xie, Z. Zhou, T. Chen, R. Liang, X. Dong, *Enhanced electrical properties of NaBi modified CaBi₂Nb₂O₉-based Aurivillius piezoceramics via structural distortion*, *Ceram. Int.* 45 (5) (2019) 5425–5430.
- [24] Z. Peng, Q. Chen, Y. Wang, D. Xin, D. Xian, J. Zhu, *Enhancement of piezoelectric properties of (LiCePr)-multidoped CaBi₂Nb₂O₉ high temperature ceramics*, *Mater. Lett.* 107 (2013) 14–16.
- [25] L. Li, H. Yuan, P. Huang, Y. Zhang, L. Sheng, P. Zheng, L. Zhang, F. Wen, W. Wu, Z. Xu, *Enhanced piezoelectricity and excellent thermal stabilities in Nb-Mg co-doped CaBi₂Ti₄O₁₅ aurivillius high curie temperature ceramics*, *Ceram. Int.* 46 (2) (2020) 2178–2184.
- [26] Z. Chen, X. Li, L. Sheng, J. Du, W. Bai, L. Li, F. Wen, P. Zheng, W. Wu, L. Zheng, *Enhanced electrical properties in A-site K/Ce and B-site W/Cr co-substituted CaBi₂Nb₂O₉ high temperature piezoelectric ceramic*, *J. Mater. Sci. Mater. Electron* 30 (2019) 11727–11734.
- [27] G. Liu, D. Wang, C. Wu, J. Wu, Q. Chen, *A realization of excellent piezoelectricity and good thermal stability in CaBi₂Nb₂O₉: pseudo phase boundary*, *J. Am. Ceram. Soc.* 102 (4) (2019) 1794–1804.
- [28] X. Tian, S. Qu, H. Ma, Z. Pei, B. Wang, *Effect of grain size on dielectric and piezoelectric properties of bismuth layer structure CaBi₂Nb₂O₉ ceramics*, *J. Mater. Sci. Mater. Electron* 27 (2016) 13309–13313.
- [29] X. Zeng, F. Cao, Z. Peng, X. Xing, *Crystal structure and electrical properties of (Li, Ce, Nd)-multidoped CaBi₂Nb₂O₉ high temperature ceramics*, *Ceram. Int.* 44 (3) (2018) 3069–3076.
- [30] W. Fan, D. Yao, S. Huang, B. Chen, B. Shen, M. Ju, Z. Dou, J. Lu, C. Wu, W. Gong, F. Yao, K. Wang, *Simultaneously achieved high piezoelectricity and resistivity in CaBi₂Nb₂O₉-based ceramics with high curie temperature*, *Ceram. Int.* 50 (7) (2024) 11667–11675.
- [31] C. Long, H. Fan, M. Li, P. Ren, Y. Cai, *A candidate for lead-free ultrahigh-temperature piezoelectrics: the excellent electro-mechanical properties of Aurivillius oxides, Ca_{1-5x}Li_{2x}Nd_{2x-3}Bi₂Nb_{2-2x}Sc_xW_xO_{9-1.5x}*, *CrystEngComm* 15 (47) (2013) 10212–10221.
- [32] H. Chen, B. Shen, J. Xu, L. Kong, J. Zhai, *Correlation between grain sizes and electrical properties of CaBi₂Nb₂O₉ piezoelectric ceramics*, *J. Am. Ceram. Soc.* 95 (11) (2012) 3514–3518.
- [33] T. Leist, J. Chen, W. Jo, E. Aulbach, J. Suffner, J. Rodel, *Temperature dependence of the piezoelectric coefficient in BiMeO₃-PbTiO₃ (Me = Fe, Sc, (Mg_{1/2}Ti_{1/2})) ceramics*, *J. Am. Ceram. Soc.* 95 (2) (2012) 711–715.
- [34] G. Liu, J. Yuan, R. Nie, L. Jiang, Z. Tan, J. Zhu, Q. Chen, *Electrical properties and thermal stability of Ce-modified Ca_{0.80}(Li_{0.5}Bi_{0.5})_{0.20}Bi₂Nb₂O₉ ceramics*, *J. Alloy. Compd.* 697 (2017) 380–387.
- [35] Y. Yang, Z. Lin, R. Li, Y. Li, W. Liu, S. Tian, K. Zhu, L. Long, *Thermal expansion coefficient of monolayer MoS₂ determined using temperature-dependent Raman spectroscopy combined with finite element simulations*, *Microstructures* 1 (1) (2021) 2021002.
- [36] C. Long, H. Fan, M. Li, Q. Li, *Crystal structure and enhanced piezoelectric properties of scandium and bismuth-excess modified aurivillius oxides, Na_{0.5-x}Bi_{2.5-x}Nb_{2-x}Sc_xO₉*, *Eur. J. Inorg. Chem.* 2013 (32) (2013) 5622–5630.
- [37] C. Qin, Z. Shen, W. Luo, F. Song, Z. Wang, Y. Li, *Microstructure related properties enhancing in Ce-doped CaBi₂Nb₂O₉ high temperature piezoelectric ceramics*, *Mater. Res. Express* 6 (10) (2019) 106308.
- [38] K. Shi, L. Peng, M. Li, Z. Zhou, K. Jiang, J. Zhang, Z. Hu, X. Dong, J. Chu, *Structural distortion, phonon behavior and electronic transition of Aurivillius layered ferroelectric CaBi₂Nb_{2-x}W_xO₉ ceramics*, *J. Alloy. Compd.* 653 (2015) 168–174.
- [39] Y. Min, H. Yu, Y. Zhang, Y. Hao, S. Chen, R. Liang, Z. Zhou, B. Yang, *Observation of La³⁺ entering (Bi₂O₂)²⁺ layer to tune tilting of NbO₆ octahedra in CaBi₂Nb₂O₉ ceramics*, *Ceram. Int.* 50 (11) (2024) 19392–19401.
- [40] I. Fujii, S. Trolier-McKinstry, *Temperature dependence of dielectric nonlinearity of BaTiO₃ ceramics*, *Microstructures* 3 (4) (2023) 2023045.
- [41] X. Tian, S. Qu, H. Du, Y. Li, Z. Xu, *Effects of (LiCe) co-substitution on the structural and electrical properties of CaBi₂Nb₂O₉ ceramics*, *Chin. Phys. B* 21 (3) (2012) 037701.
- [42] C. Zhu, A. Chen, Y. Liu, P. Zheng, W. Bai, Q. Fan, L. Zheng, Y. Zhang, *Realizing enhanced electrical properties of CaBi₂Nb₂O₉-based high-temperature piezoceramics by constructing a pseudophase boundary*, *ACS Appl. Electron Mater.* 4 (7) (2022) 3598–3605.
- [43] N. Chen, F. Wang, X. Li, Y. Xie, J. Xi, S. Guan, W. Shi, H. Chen, J. Xing, J. Zhu, *Improving the piezoelectric properties of CBN-based ceramic by a Ce ion*, *J. Am. Ceram. Soc.* 105 (10) (2022) 6207–6216.
- [44] G. Liu, C. Wu, Y. Chen, D. Liang, B. Wang, J. Wu, Q. Chen, *Enhanced ferroelectricity of CaBi₂Nb₂O₉-based high-temperature piezoceramics by pseudo-tetragonal distortion*, *Ceram. Int.* 44 (6) (2018) 5880–5885.
- [45] S. Katayama, Y. Noguchi, M. Miyayama, *3D domain structure in Bi₄Ti₃O₁₂ crystals observed by using piezoresponse force microscopy*, *Adv. Mater.* 19 (18) (2007) 2552–2555.